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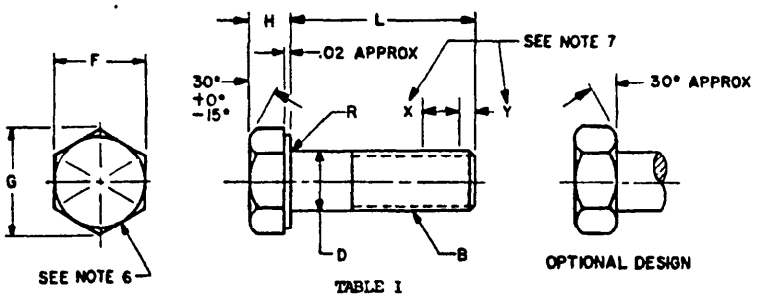


TABLE I

NOMINAL SIZE		1/4	5/16	3/8	7/16	1/2	9/16	5/8
B THREAD UNF-2A		28	24	24	20	20	18	18
D BODY DIAMETER	MAX	.2500	.3125	.3750	.4375	.5000	.5625	.6250
	MIN	.2450	.3065	.3690	.4305	.4930	.5545	.6170
F WIDTH ACROSS FLATS	MAX	.4375	.5000	.5625	.6250	.7500	.8125	.9375
	MIN	.4280	.4890	.5510	.6120	.7360	.7980	.9220
G WIDTH ACROSS CORNERS	MAX	.505	.577	.650	.722	.866	.938	1.083
	MIN	.488	.557	.628	.698	.840	.910	1.051
H HEAD HEIGHT	MAX	.163	.211	.243	.291	.323	.371	.403
	MIN	.150	.195	.226	.272	.302	.348	.378
R RADIUS	MAX	.025	.025	.025	.025	.025	.045	.045
	MIN	.015	.015	.015	.015	.015	.020	.020
* MIN TENSILE STRENGTH - LBS		5,450	8,700	13,200	17,800	24,000	30,400	38,400

L LENGTH	TOLERANCE	DASH NO.	DASH NO.	DASH NO.	DASH** NO.	DASH NO.	DASH** NO.	DASH NO.
.375		1	27	53				
.438		2	28	54	79			
.500		3	29	55	80	104		
.562	+0	4	30	56	81	105	130	
.625	-.031	5	31	57	82	106	131	155
.750		6	32	58	83	107	132	156
.875		7	33	59	84	108	133	157
1.000		8	34	60	85	109	134	158
1.125		9	35	61	86	110	135	159
1.250		10	36	62	87	111	136	160
1.375	+0	11	37	63	88	112	137	161
1.500	-.062	12	38	64	89	113	138	162
1.750		13	39	65	90	114	139	163
2.000		14	40	66	91	115	140	164
2.250		15	41	67	92	116	141	165
2.500		16	42	68	93	117	142	166
2.750		17	43	69	94	118	143	167
3.000		18	44	70	95	119	144	168
3.250		19	45	71	96	120	145	169
3.500	+0	20	46	72	97	121	146	170
3.750	-.094	21	47	73	98	122	147	171
4.000		22	48	74	99	122	148	172
4.250		21	49	75	100	124	149	173
4.500		24	50	76	101	125	150	174
4.750		25	51	77	102	126	151	175
5.000		26	52	78	103	127	152	176
5.500						128	153	177
6.000						129	154	178

NOTE: FOR NOTES AND ADDITIONAL SIZES SEE SHEET 2.

CROSS REFERENCE OF PART NUMBERS

THE SCREWS COVERED BY DASH NUMBERS GIVEN IN MS35302, MS35304 AND MS35306 ARE CANCELLED AFTER THE DATES INDICATED ON THE CANCELLED DOCUMENTS. USE DASH NUMBERS USED IN THIS STANDARD. THE CANCELLED SCREWS CANNOT ALWAYS REPLACE THE NEW SCREWS AND SHOULD BE USED UNTIL EXISTING STOCKS ARE DEPLETED. USE ONLY THE NEW SCREWS FOR DESIGN AND REPLACEMENT. DASH NUMBERS REPRESENTING OLD AND NEW SCREWS ARE IDENTICAL, MAKING IT UNNECESSARY TO PREPARE AN INTERCHANGEABILITY TABLE.

(F) DENOTES CHANGES

APPROVED 6 MAY 63 REVISED (A) 1 AUG 67 (B) 14 OCT 68 (C) 1 MAY 1969 (D) 22 OCT 1970 (E) 21 JUNE 1978 (F) 4 NOV 88

P.A. AR
Other Cust 99

INTERNATIONAL INTEREST

TITLE: SCREW, CAP, HEXAGON HEAD (FINISHED HEXAGON BOLT), ALLOY STEEL, GRADE 8, ZINC COATED, UNF-2A, PLAIN AND SELF-LOCKING (F)

MILITARY STANDARD
MS90727

PROCUREMENT SPECIFICATION: FF-S-85

SUPERSEDES: SEE NOTE 12

SHEET 1 OF 5

USER ACTIVITIES:
ME MU PA CL WV OB MC NSA

REVIEWER ACTIVITIES:
AT AV MI EA

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TABLE II

NOMINAL SIZE		3/4	7/8	1	1-1/8	1-1/4	1-3/8	1-1/2
B THREAD UNF-2A		16	14	12	12	12	12	12
D BODY DIAMETER	MAX	.7500	.8750	1.0000	1.1250	1.2500	1.3750	1.5000
	MIN	.7410	.8660	.9900	1.1140	1.2390	1.3630	1.4880
F WIDTH ACROSS FLATS	MAX	1.1250	1.3125	1.5000	1.6875	1.8750	2.0625	2.2500
	MIN	1.1000	1.2850	1.4690	1.6310	1.8120	1.9940	2.1750
G WIDTH ACROSS CORNERS	MAX	1.299	1.516	1.732	1.949	2.165	2.382	2.598
	MIN	1.254	1.465	1.675	1.859	2.066	2.273	2.480
H HEAD HEIGHT	MAX	.483	.563	.627	.718	.813	.878	.974
	MIN	.455	.531	.591	.658	.749	.810	.902
R RADIUS	MAX	.045	.065	.095	.095	.095	.095	.095
	MIN	.020	.040	.060	.060	.060	.060	.060
* MIN TENSILE STRENGTH - LBS		56,000	76,400	99,400	128,400	161,000	197,200	237,200

L LENGTH	TOLERANCE SIZE .750	TOLERANCE SIZE .875 AND OVER	DASH NO.	DASH** NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.
.750			179						
.875	+0	+0	180	202					
1.000	-.031	-.062	181	201	224				
1.125			182	204	225	245			
1.250	+0	+0	183	205	226	246	265		
1.375	-.062	-.125	184	206	227	247	266	284	
1.500			185	207	228	248	267	285	302
1.750			186	208	229	249	268	286	303
2.000			187	209	230	250	269	287	304
2.250			188	210	231	251	270	288	305
2.500			189	211	232	252	271	289	306
2.750			190	212	233	253	272	290	307
3.000			191	213	234	254	273	291	308
3.250			192	214	235	255	274	292	309
3.500	+0	+0	193	215	236	256	275	293	310
3.750	-.094	-.188	194	216	237	257	276	294	311
4.000			195	217	238	258	277	295	312
4.250			196	218	239	259	278	296	313
4.500			197	219	240	260	279	297	314
4.750			198	220	241	261	280	298	315
5.000			199	221	242	262	281	299	316
5.500			200	222	243	263	282	300	317
6.000			201	223	244	264	283	301	318

NOTES:

1. MATERIAL: ALLOY STEEL, GRADE 8, IN ACCORDANCE WITH PROCUREMENT SPECIFICATION.
2. (F) PROTECTIVE COATING: ZINC COATING IN ACCORDANCE WITH PROCUREMENT SPECIFICATION.
3. THREADS: THE THREADS SHALL BE IN ACCORDANCE WITH SCREW-THREAD STANDARDS FOR FEDERAL SERVICES, HANDBOOK H28.
4. THREAD LENGTH: MINIMUM THREAD LENGTH SHALL BE TWICE THE DIAMETER PLUS 1/4 INCH. THE TOLERANCE SHALL BE PLUS 3/16 INCH OR 2-1/2 THREADS, WHICHEVER IS GREATER. ON SCREWS THAT ARE TOO SHORT FOR MINIMUM THREAD LENGTHS, THE DISTANCE FROM THE BEARING SURFACE OF THE HEAD TO THE FIRST COMPLETE THREAD SHALL NOT EXCEED THE LENGTH OF 2-1/2 THREADS FOR SIZES UP TO AND INCLUDING 1 INCH AND 3-1/2 THREADS FOR SIZES LARGER THAN 1 INCH.
5. DIMENSIONS: ALL DIMENSIONS ARE IN INCHES, UNLESS OTHERWISE SPECIFIED.
6. HEAD MARKING: HEADS SHALL BE MARKED IN ACCORDANCE WITH MIL-HDBK-131. RAISED MARKINGS ARE PREFERRED, STAMPINGS ARE ACCEPTABLE.

APPROVED 6 MAY 63 REVISED (A) 1 AUG 67 (B) 14 OCT 68 (C) MAY 1969 (D) 22 OCT 1970 (F) FOR CHANGES SEE SHEETS 1,2,3,4,AND 5

USER ACTIVITIES: ME MU PA CL WV OB MC NSA

REVIEWER ACTIVITIES: AT AV MI EA

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P.A.	INTERNATIONAL INTEREST	TITLE	
Other Cust AR 99		SCREW, CAP, HEXAGON HEAD (FINISHED HEXAGON BOLT), ALLOY STEEL, GRADE 8, ZINC COATED, UNF-2A, PLAIN AND SELF-LOCKING (F)	MILITARY STANDARD MS90727
PROCUREMENT SPECIFICATION FF-S-85	SUPERSEDES SEE NOTE 12		SHEET 2 OF 5

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7. SELF-LOCKING ELEMENT:

(a) THE SELF-LOCKING ELEMENT SHALL BE A PATCH TYPE OR LONGITUDINAL STRIP IN ACCORDANCE WITH MIL-F-18240.

(b) FOR "X" AND "Y" DIMENSIONS AND DESIGN AND USAGE LIMITATIONS, SEE MS15981.

(c) MAXIMUM RING GAGE DIAMETER THAT LOCKING REGION OF SCREW MUST PASS THRU FREELY OR WITH FINGER PRESSURE SHALL BE THE NOMINAL SIZE PLUS 0.010.

8. PART NUMBER: THE MS PART NUMBER CONSISTS OF THE MS NUMBER, PLUS THE DASH NUMBER. FOR SCREWS WITH SELF-LOCKING ELEMENT ADD THE LETTER "L" AFTER THE DASH NUMBER.

EXAMPLE: MS90727-1 PLAIN SCREW.
MS90727-1L SELF-LOCKING SCREW.

*9. BASED ON MINIMUM ULTIMATE TENSILE STRENGTH OF 150,000 p.s.i. TENSILE STRENGTHS ARE CALCULATED BY THE STRESS AREAS INDICATED IN SCREW-THREAD STANDARDS FOR FEDERAL SERVICES, HANDBOOK H28.

10. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID.

11. FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN.

12. SUPERSEDES: MS35302 IN PART, MS35304 AND MS35306, BCAX8, BCVX7 IN PART, BQLX4, BUBX17, 17.1, 18, 18.1, 19 AND 19.1.

*13. INACTIVE FOR NEW DESIGN AFTER 21 JUNE 1976.

(F)14. THIS MILITARY STANDARD IS NOT INTENDED FOR USE IN AIRBORNE APPLICATIONS.

APPROVED 6 MAY 63 REVISED (A) 1 AUG 67 (B) 14 OCT 68 (C) 1 MAY 1969 (D) 22 OCT 1970 (F) FOR CHANGES SEE SHEETS 1,2,3,4,AND 5

USER ACTIVITIES:
ME MU PA CL WV OB MC NSA

REVIEWER ACTIVITIES:
AT AV MI EA

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P.A. AR	INTERNATIONAL INTEREST	TITLE	MILITARY STANDARD
Other Cust 99		SCREW, CAP, HEXAGON HEAD (FINISHED HEXAGON BOLT), ALLOY STEEL, GRADE 8, ZINC COATED, UNF-2A, PLAIN AND SELF-LOCKING (F)	MS90727
PROCUREMENT SPECIFICATION FF-S-85	SUPERSEDES SEE NOTE 12		SHEET 3 OF 5

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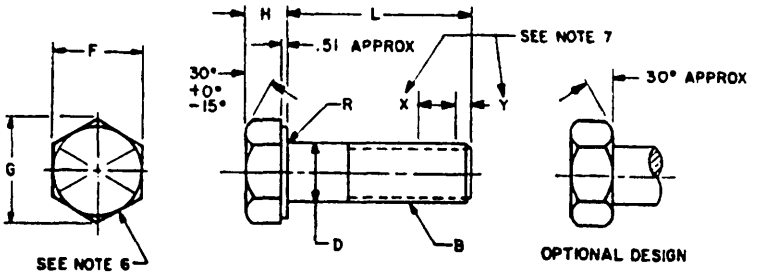


TABLE III
METRIC TRANSLATION OF TABLE I

NOMINAL SIZE	MM	6.350	7.938	9.525	11.112	12.700	14.288	15.875
B THREAD PITCH		0.907	1.058	1.058	1.270	1.270	1.411	1.411
D BODY DIAMETER	MAX	6.350	7.938	9.525	11.112	12.700	14.288	15.875
	MIN	6.223	7.785	9.373	10.935	12.522	14.084	15.672
F WIDTH ACROSS FLATS	MAX	11.112	12.700	14.288	15.875	19.050	20.638	23.812
	MIN	10.871	12.421	13.995	15.545	18.694	20.269	23.419
G WIDTH ACROSS CORNERS	MAX	12.827	14.656	16.510	18.339	21.996	23.825	27.508
	MIN	12.395	14.148	15.951	17.729	21.336	23.114	26.695
H HEAD HEIGHT	MAX	4.140	5.359	6.172	7.391	8.204	9.423	10.236
	MIN	3.810	4.953	5.740	6.909	7.671	8.839	9.601
R RADIUS	MAX	.635	.635	.635	.635	.635	1.143	1.143
	MIN	.381	.381	.381	.381	.381	.508	.508
* MIN TENSILE STRENGTH-KN		24.525	39.150	59.400	80.100	108.000	136.800	172.800

L LENGTH	TOLERANCE	DASH NO.	DASH NO.	DASH NO.	DASH NO.**	DASH NO.	DASH NO.**	DASH NO.
9.525		1	27	53				
11.125		2	28	54	79			
12.700		3	29	55	80	104		
14.275	+0 -.787	4	30	56	81	105	130	
15.875		5	31	57	82	106	131	155
19.050		6	32	58	83	107	132	156
22.225		7	33	59	84	108	133	157
25.400		8	34	60	85	109	134	158
28.575		9	35	61	86	110	135	159
31.750	+0	10	36	62	87	111	136	160
34.925	-1.575	11	37	63	88	112	137	161
38.100		12	38	64	89	113	138	162
44.450		13	39	65	90	114	139	163
50.800		14	40	66	91	115	140	164
57.150		15	41	67	92	116	141	165
63.500		16	42	68	93	117	142	166
69.850		17	43	69	94	118	143	167
76.200		18	44	70	95	119	144	168
82.550		19	45	71	96	120	145	169
88.900	+0	20	46	72	97	121	146	170
95.250	-2.388	21	47	73	98	122	147	171
101.600		22	48	74	99	123	148	172
107.950		23	49	75	100	124	149	173
114.300		24	50	76	101	125	150	174
120.650		25	51	77	102	126	151	175
127.000		26	52	78	103	127	152	176
139.700						128	153	177
152.400						129	154	178

NOTE:

1. ALL DIMENSIONS ARE IN MILLIMETERS (mm), UNLESS OTHERWISE SPECIFIED AND ARE BASED ON AN APPROXIMATE CONVERSION FACTOR OF 25.4 mm EQUAL ONE INCH.
2. FOR ADDITIONAL SIZES SEE SHEET 4.

**3. INACTIVE FOR NEW DESIGN AFTER 21 JUNE 1976.

P.A.	INTERNATIONAL INTEREST	TITLE	MILITARY STANDARD
AR		SCREW, CAP, HEXAGON HEAD (FINISHED HEXAGON BOLT), ALLOY STEEL, GRADE 8, ZINC COATED, UNF-2A, PLAIN AND SELF-LOCKING (F)	MS90727
Other Cust 99			
PROCUREMENT SPECIFICATION: FF-S-85	SUPERSEDES: SEE NOTE 12		SHEET 4 OF 5

APPROVED 6 MAY 63 REVISED (A) 1 AUG 1967 (B) 14 OCT 1968 (C) (D) 22 OCT 1970 (F) FOR CHANGES SEE SHEETS 1,2,3,4,AND 5
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USER ACTIVITIES:
ME MU PA CL WV OB MC NSA

REVIEWER ACTIVITIES:
AT AV MI EA

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TABLE IV
METRIC TRANSLATION OF TABLE II

	NOMINAL SIZE	MM	19.050	22.225	25.400	28.575	31.750	34.925	38.100
B	THREAD PITCH		1.588	1.814	2.117	2.117	2.117	2.117	2.117
D	BODY DIAMETER	MAX	19.050	22.225	25.400	28.575	31.750	34.925	38.100
		MIN	18.821	21.996	25.146	28.296	31.471	34.620	37.795
F	WIDTH ACROSS FLATS	MAX	28.575	33.338	38.100	42.862	47.625	52.388	57.150
		MIN	27.940	32.639	37.313	41.427	46.025	50.648	55.245
G	WIDTH ACROSS CORNERS	MAX	32.995	38.506	43.993	49.505	54.991	60.503	65.989
		MIN	31.852	37.211	42.545	47.219	52.476	57.734	62.992
H	HEAD HEIGHT	MAX	12.268	14.300	15.926	18.237	20.650	22.301	24.740
		MIN	11.557	13.487	15.011	16.713	19.025	20.574	22.911
R	RADIUS	MAX	1.143	1.651	2.413	2.413	2.413	2.413	2.413
		MIN	.508	1.016	1.524	1.524	1.524	1.524	1.524
**MIN TENSILE STRENGTH-KN			252.000	343.800	447.300	577.800	724.500	887.400	1067.400

L LENGTH	TOLERANCE: SIZE 19.050	TOLERANCE: SIZE 22.225 AND OVER	DASH NO.	DASH** NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.
19.050	+0	+0	179						
22.225	-.787	-1.575	180	202					
25.400			181	203	224				
28.575			182	204	225	245			
31.750	+0	+0	183	205	226	246	265		
34.925	-1.575	-3.175	184	206	227	247	266	284	
38.100			185	207	228	248	267	285	302
44.450			186	208	229	249	268	286	303
50.800			187	209	230	250	269	287	304
57.150			188	210	231	251	270	288	305
63.500			189	211	232	252	271	289	306
69.850			190	212	233	253	272	290	307
76.200			191	213	234	254	273	291	308
82.550			192	214	235	255	274	292	309
88.900	+0	+0	193	215	236	256	275	293	310
95.250	-2.388	-4.775	194	216	237	257	276	294	311
101.600			195	217	238	258	277	295	312
107.950			196	218	239	259	278	296	313
114.300			197	219	240	260	279	297	314
120.650			198	220	241	261	280	298	315
127.000			199	221	242	262	281	299	316
139.700			200	222	243	263	282	300	317
152.400			201	223	244	264	283	301	318

NOTE:

1. ALL DIMENSIONS ARE IN MILLIMETERS (mm), UNLESS OTHERWISE SPECIFIED AND ARE BASED ON AN APPROXIMATE CONVERSION FACTOR OF 25.4 mm EQUAL ONE INCH.

2. THREAD LENGTH: MINIMUM THREAD LENGTH SHALL BE TWICE THE DIAMETER PLUS 6.350 mm. THE TOLERANCE SHALL BE PLUS 4.775 mm OR 2-1/2 THREADS, WHICHEVER IS GREATER. ON SCREWS THAT ARE TOO SHORT FOR MINIMUM THREAD LENGTHS, THE DISTANCE FROM THE BEARING SURFACE OF THE HEAD TO THE FIRST COMPLETE THREAD SHALL NOT EXCEED THE LENGTH OF 2-1/2 THREADS FOR SIZES UP TO AND INCLUDING 25.400 mm AND 3-1/2 THREADS FOR SIZES LARGER THAN 25.400 mm.

**3. INACTIVE FOR NEW DESIGN AFTER 21 JUNE 1976.

APPROVED 6 MAY 63 REVISED (A) 1 AUG 1967 (B) 14 OCT 1968 (C) 1 MAY 1969 (D) 22 OCT 1970 (F) FOR CHANGES SEE SHEETS 1,2,3,4,AND 5

USER ACTIVITIES:
ME MU PA CL WV OB MC NSA

REVIEWER ACTIVITIES:
AT AV MI EA

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P.A.	INTERNATIONAL INTEREST	TITLE	MILITARY STANDARD
AR		SCREW, CAP, HEXAGON HEAD (FINISHED HEXAGON BOLT), ALLOY STEEL, GRADE 8, ZINC COATED, UNF-2A, PLAIN AND SELF-LOCKING (F)	MS90727
Other Cust 99			
PROCUREMENT SPECIFICATION: FF-S-85	SUPERSEDES: SEE NOTE 12		SHEET 5 OF 5

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